Derivatives Using the Power Rule

Calculus Worksheet · Grade 11–12 / College Intro Calculus

Name: ________________________________ Date: ____________________

Learning Objectives

- Apply the Power Rule to find derivatives of polynomial and rational functions
- Convert radical and rational expressions into exponential form before differentiating
- Simplify derivative results using algebraic manipulation and negative/fractional exponents

Problems

1. Find the derivative of the function using the Power Rule.

$$y = x^5$$

2. Find the derivative of the function using the Power Rule.

$$y = 4x^3 - 7x^2 + 2x - 9$$

3. Find the derivative of the function. Rewrite the fraction by separating constants before applying the Power Rule.

$$y = \frac{x^4}{4} + \frac{x^3}{3} - x$$

4. Convert the square root to a fractional exponent, then find the derivative.

$$f(x) = \sqrt{x}$$

5. Convert the cube root to a fractional exponent, then find the derivative.

$$g(x) = 3\sqrt[3]{x^2}$$

Free to print & share (CC BY 4.0)

6. Rewrite the expression by dividing each term in the numerator by x, then find the derivative using the Power Rule.

$$f(x) = \frac{x^3 + 5x^2 - 2}{x}$$

7. Find the derivative of the function, which includes an exponential term. Recall that the derivative of e to the x equals e to the x.

$$h(x) = 6x^4 - 5e^x + 3$$

8. Convert all radical and negative-power terms into fractional exponents, then find the derivative and simplify.

$$g(x) = \frac{5}{x^{3/5}} - 7e^x - 4\sqrt[3]{x}$$

9. Rewrite the function by distributing x to the negative 1 to eliminate the denominator, then find the derivative.

$$f(x) = \frac{x^3 - 4x + 6}{x^2}$$

10. Find the derivative of the function by first rewriting all terms using positive or negative integer and fractional exponents, then apply the Power Rule to each term and simplify fully.

$$y = \frac{x^3}{3} + \frac{x^2}{2} + x - \frac{3}{\sqrt{x}} + 2e^x$$

Free to print & share (CC BY 4.0)

Derivatives Using the Power Rule — Answer Key

Calculus Worksheet · Grade 11–12 / College Intro Calculus

Answer Key

1. Answer: $y' = 5x^4$

- Apply the Power Rule: if $y = x^n$, then $y' = n \cdot x^{(n-1)}$
- Bring down the exponent: $y' = 5 \cdot x^{(5-1)}$
- Simplify: $y' = 5x^4$

2. Answer: $y' = 12x^2 - 14x + 2$

- Differentiate each term separately using the Power Rule
- $d/dx(4x^3) = 12x^2$
- $d/dx(-7x^2) = -14x$
- $d/dx(2x) = 2$
- $d/dx(-9) = 0$ (constant rule)
- Combine: $y' = 12x^2 - 14x + 2$

3. Answer: $y' = x^3 + x^2 - 1$

- Rewrite as $y = (1/4)x^4 + (1/3)x^3 - x$
- $d/dx[(1/4)x^4] = (1/4) \cdot 4x^3 = x^3$
- $d/dx[(1/3)x^3] = (1/3) \cdot 3x^2 = x^2$
- $d/dx[-x] = -1$
- Combine: $y' = x^3 + x^2 - 1$

4. Answer: $f'(x) = 1/(2\sqrt{x})$ or $(1/2)x^{(-1/2)}$

- Rewrite: $f(x) = x^{(1/2)}$
- Apply the Power Rule: $f'(x) = (1/2)x^{(1/2 - 1)}$
- Simplify the exponent: $1/2 - 1 = -1/2$
- $f'(x) = (1/2)x^{(-1/2)} = 1/(2\sqrt{x})$

5. Answer: $g'(x) = 2x^{(-1/3)}$ or 2/■x

- Rewrite: $g(x) = 3x^{(2/3)}$
- Apply the Power Rule: $g'(x) = 3 \cdot (2/3)x^{(2/3 - 1)}$
- Simplify the coefficient: $3 \cdot (2/3) = 2$
- Simplify the exponent: $2/3 - 1 = 2/3 - 3/3 = -1/3$
- $g'(x) = 2x^{(-1/3)}$ = 2/■x

6. Answer: $f'(x) = 2x + 5 + 2x^{(-2)}$

- Divide each term by x: $f(x) = x^2 + 5x - 2x^{(-1)}$
- Apply the Power Rule to each term
- $d/dx[x^2] = 2x$
- $d/dx[5x] = 5$
- $d/dx[-2x^{(-1)}] = -2 \cdot (-1)x^{(-2)} = 2x^{(-2)}$

Free to print & share (CC BY 4.0)

- Combine: f'(x) = 2x + 5 + 2x^(–2)

7. Answer: h'(x) = 24x^3 - 5e^x
- d/dx[6x^4] = 24x^3
- d/dx[–5e^x] = –5e^x (since d/dx[e^x] = e^x)
- d/dx[3] = 0
- Combine: h'(x) = 24x^3 – 5e^x

8. Answer: g'(x) = –3x^(–8/5) – 7e^x – (4/3)x^(–2/3)
- Rewrite: g(x) = 5x^(–3/5) – 7e^x – 4x^(1/3)
- d/dx[5x^(–3/5)] = 5·(–3/5)x^(–3/5 – 1) = –3x^(–8/5)
- d/dx[–7e^x] = –7e^x
- d/dx[–4x^(1/3)] = –4·(1/3)x^(1/3 – 1) = –(4/3)x^(–2/3)
- Combine: g'(x) = –3x^(–8/5) – 7e^x – (4/3)x^(–2/3)

9. Answer: f'(x) = 1 + 4x^(–3) – 12x^(–4)
- Divide each term by x^2: f(x) = x – 4x^(–1) + 6x^(–2)
- d/dx[x] = 1
- d/dx[–4x^(–1)] = –4·(–1)x^(–2) = 4x^(–2)
- Wait — recheck: d/dx[–4x^(–1)] = 4x^(–2)
- d/dx[6x^(–2)] = 6·(–2)x^(–3) = –12x^(–3)
- Combine: f'(x) = 1 + 4x^(–2) – 12x^(–3)

10. Answer: y' = x^2 + x + 1 + (3/2)x^(–3/2) + 2e^x
- Rewrite: y = (1/3)x^3 + (1/2)x^2 + x – 3x^(–1/2) + 2e^x
- d/dx[(1/3)x^3] = (1/3)·3x^2 = x^2
- d/dx[(1/2)x^2] = (1/2)·2x = x
- d/dx[x] = 1
- d/dx[–3x^(–1/2)] = –3·(–1/2)x^(–3/2) = (3/2)x^(–3/2)
- d/dx[2e^x] = 2e^x
- Combine: y' = x^2 + x + 1 + (3/2)x^(–3/2) + 2e^x

Free to print & share (CC BY 4.0)